

7.1: Inverse Functions

- Find the inverse, $f^{-1}(x)$, of a function on a given interval and its domain.
 - Directly; **Problems 15-19 and 25-27.**
- Verify the relationship, i.e., $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$; **Problems 15-19.**
- Graph the inverse function.
 - From a function; **Problems 25-27.**
 - From a graph; **Problems 29-30 (these were not assigned).**
- Find the derivative of an inverse function using $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}$ where $y_0 = x_0$; **Problems 35-37.**

7.2: Differentiation and Integrating the Logarithmic and Exponential Functions

- Find the derivatives of functions with $\ln x$, $\ln(u(x))$, e^x , and $e^{u(x)}$; **Problems 15-19 and 25-27.**
 - You may have to use the product rule, quotient rule, and/or chain rule.
- Find the indefinite and definite integrals of functions with e^x , $e^{u(x)}$, $\frac{1}{x}$, and $\frac{1}{u(x)}$ (not $\ln x$); **Problems 31-35, 37.**
 - You may have to use substitution.
- You will NOT be given problems where you have to use logarithmic differentiation, $\frac{d}{dx} \ln f(x) = f'(x) \frac{1}{f(x)}$ (Problems 39-41).

7.3: Logarithmic and Exponential Functions with other bases.

- Understanding log functions; **Problems 11-15.**
- Finding the derivative and indefinite integrals of logarithmic and exponential functions with bases other than e .
 - Derivatives of b^x ; **Problems 21-25.**
 - Indefinite Integrals with general bases; **Problems 30-33 (indefinite only).**
- Find a derivative using the general power rule. **Problems 35, 36, 42.**
 - The results of section 7.3 allows the power rule (early stated for rational powers) to be extended to include all real numbers.
- Derivative and integral Exam problems from this section will be straight forward application of the formulas. There will probably be at most one problem taken from each item above.

8.1 Integration by Parts.

- Find the indefinite integral of a function using integration by parts; **Problems 7-19.**
 - You may have to use substitution. You will NOT have to use repeated integration by parts (Problems 23-28).
- Find the definite integral of a function using integration by parts; **Problems 29-33.**
 - Some of the HW problems may require trigonometric substitutions found in earlier problems (e.g., 35). You will NOT have to use trigonometric substitutions on the exam; we will cover this in 8.2 and 8.3.

Some general notes.

- You will be given the necessary formulas from section 7.1, 7.2, 7.3, and 8.1 (as they appear in the text). You will also be given the product, quotient, chain, and trigonometric rules.
- Of the four sections, 7.3 will be covered the least, however be prepared to use the formulas from that section.
- There will be 15-20 problems including some “easy” problems involving properties of logs and inverse functions.